JLCPCB Order Errors

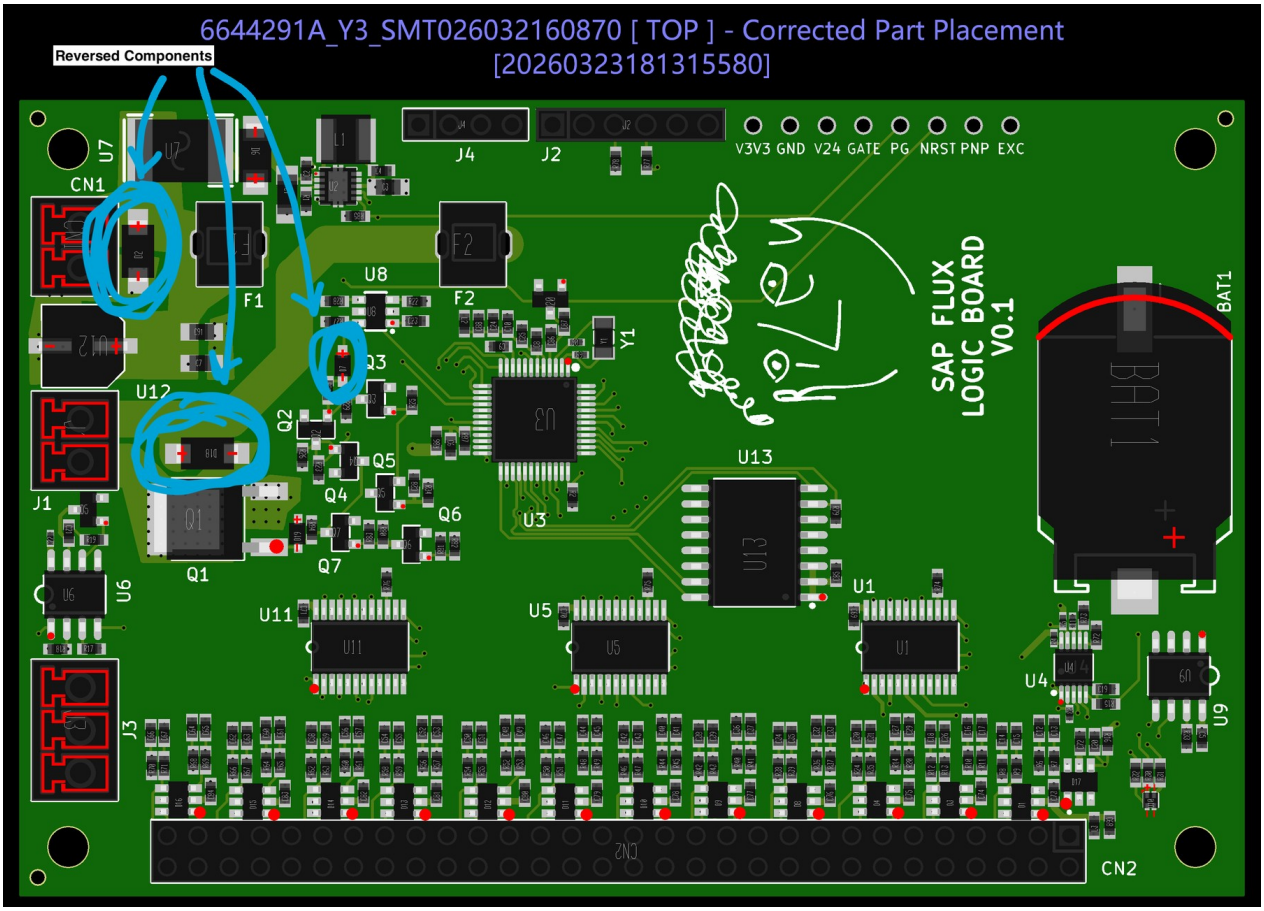
Riley Leff

June 2026

Order: `SMT026032160870` / `W2026032010337545`

Hi, I found that my PCBA had three two-terminal polarized diodes installed with the cathode stripe positioned opposite the one specified in my files and confirmed by JLC: `D2`, `D7`, and `D18`.

This is the component orientation I confirmed before assembly, see the blue annotations for the three components of interest:



Annotated orientation image showing D2, D7, and D18

Overview

Ref	Part	Expected from confirmed orientation	Observed on assembled board	Evidence
`D2`	`SMAJ33A`, `C693627`	cathode/stripe on pad 1, `V24_RAW`; anode on pad 2, `GND`	stripe was on `GND` side; part failed short immediately on 24 V input	post-failure photo, short cleared after D2 removal

Ref	Part	Expected from confirmed orientation	Observed on assembled board	Evidence
D7	1N4148WS, C2128	cathode/stripe on pad 1, NRST; anode on pad 2, d_iso_anode_node	stripe on bottom pad; top pad continuity to NRST, bottom pad no continuity to NRST	microscope photo plus continuity check
D18	SS56, C18199169	cathode/stripe on pad 1, heater_hi; anode on pad 2, heater_lo	stripe on right side, closer to Q2, which is the heater_lo side	microscope photo

Details And Photos

D2

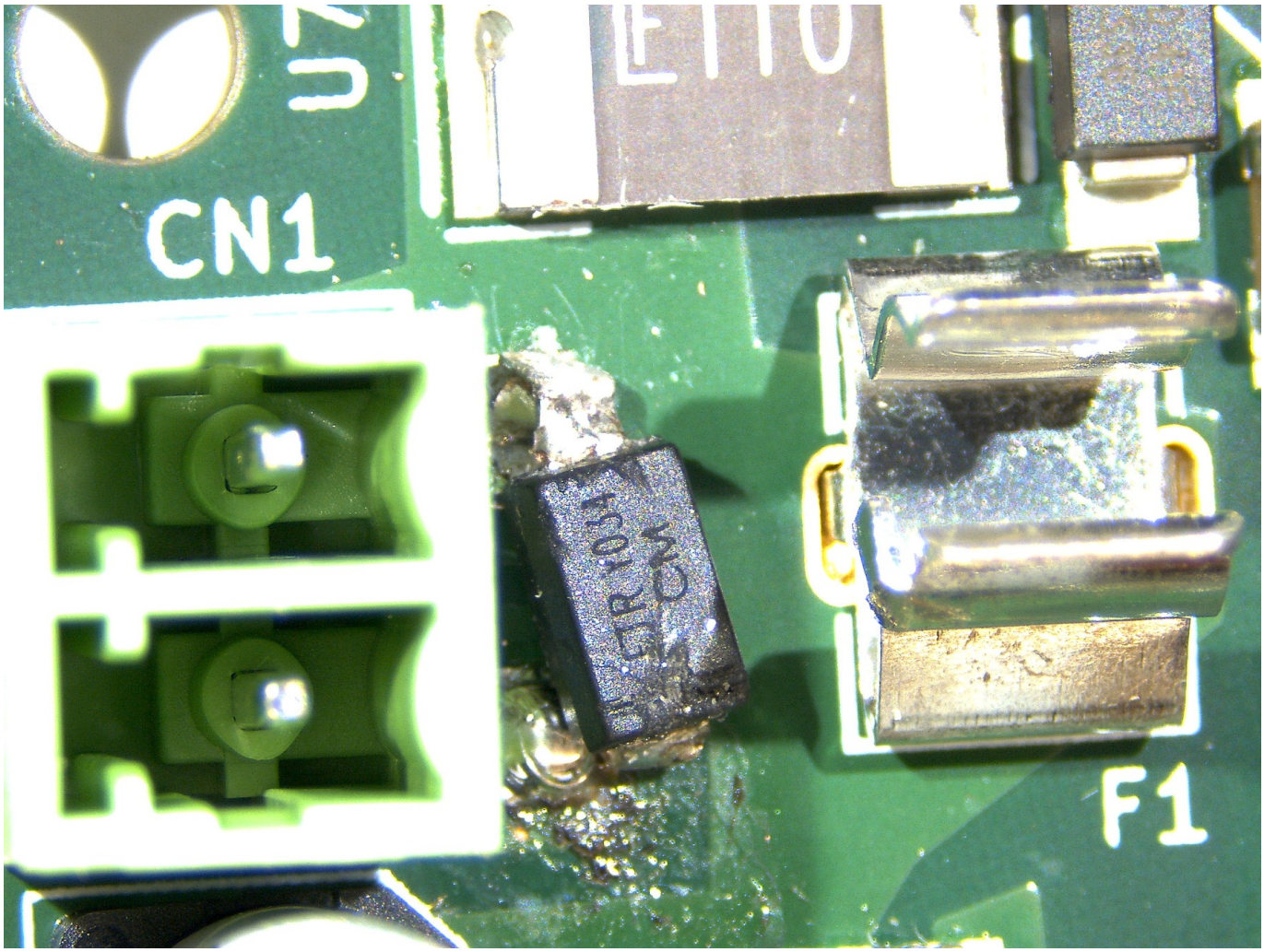
This reversed diode caused an immediate short when I plugged in power for the first time and blew itself out. I removed the part to eliminate the short and continue testing the board, but replaced it for the photo below.

Submitted orientation:

- D2 pad 1 (upward, in this image) is V24_RAW.
- D2 pad 2 is GND.
- The confirmed orientation above maps pin 1 to cathode and pin 2 to anode.

Observed:

- The delivered board had the D2 cathode stripe on the GND side.



D2 microscope photo after failure/removal

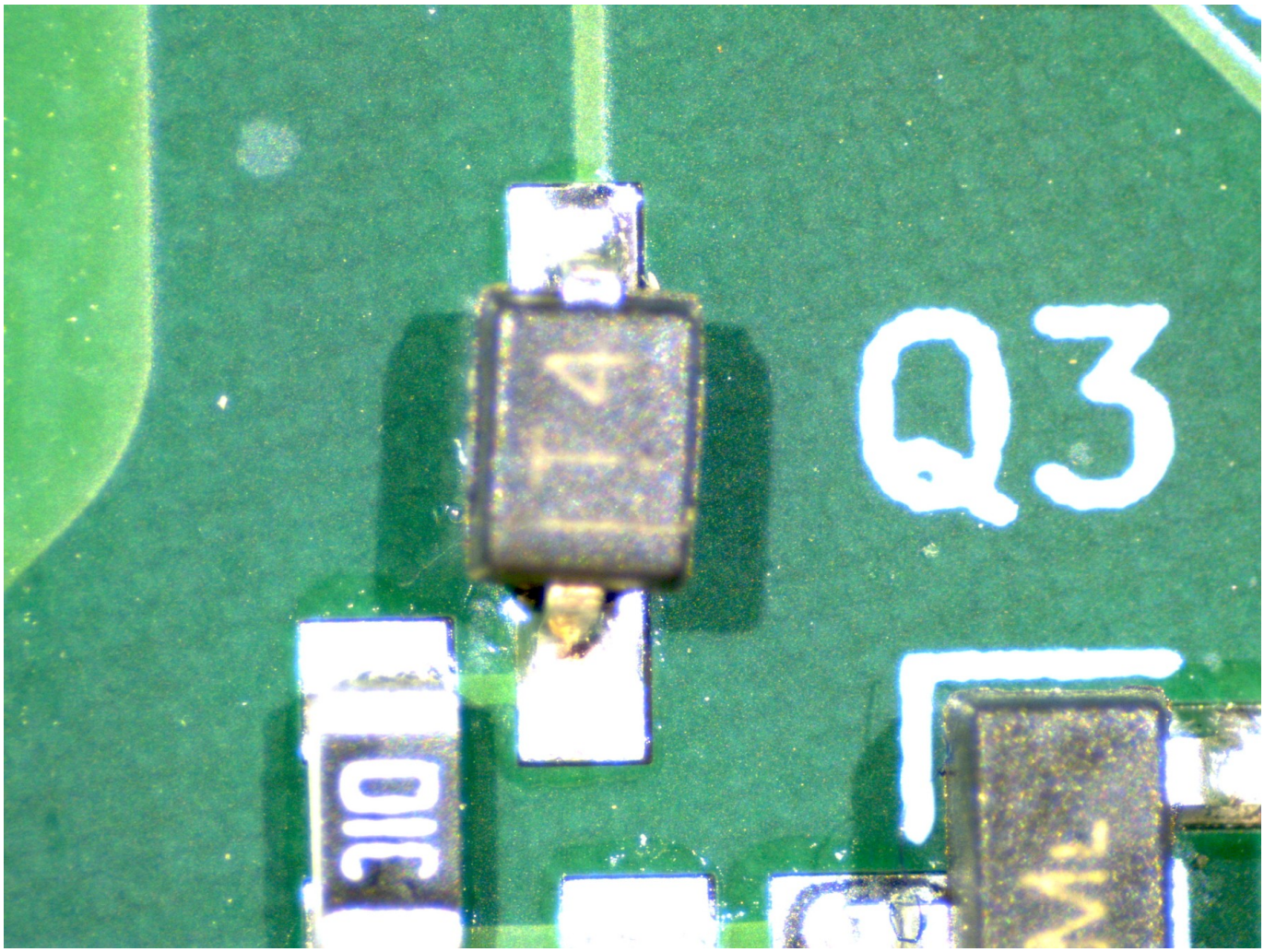
D7

Submitted orientation:

- `D7` pad 1 (upward, in this image) is `NRST`.
- `D7` pad 2 is `d_iso_anode_node`.
- The confirmed orientation maps pin 1 to cathode and pin 2 to anode.

Observed:

- The stripe is oriented towards the bottom pad.
- Continuity check: top pad is continuous with the `NRST` test pad.
- Continuity check: bottom pad is not continuous with the `NRST` test pad.



D7 microscope photo

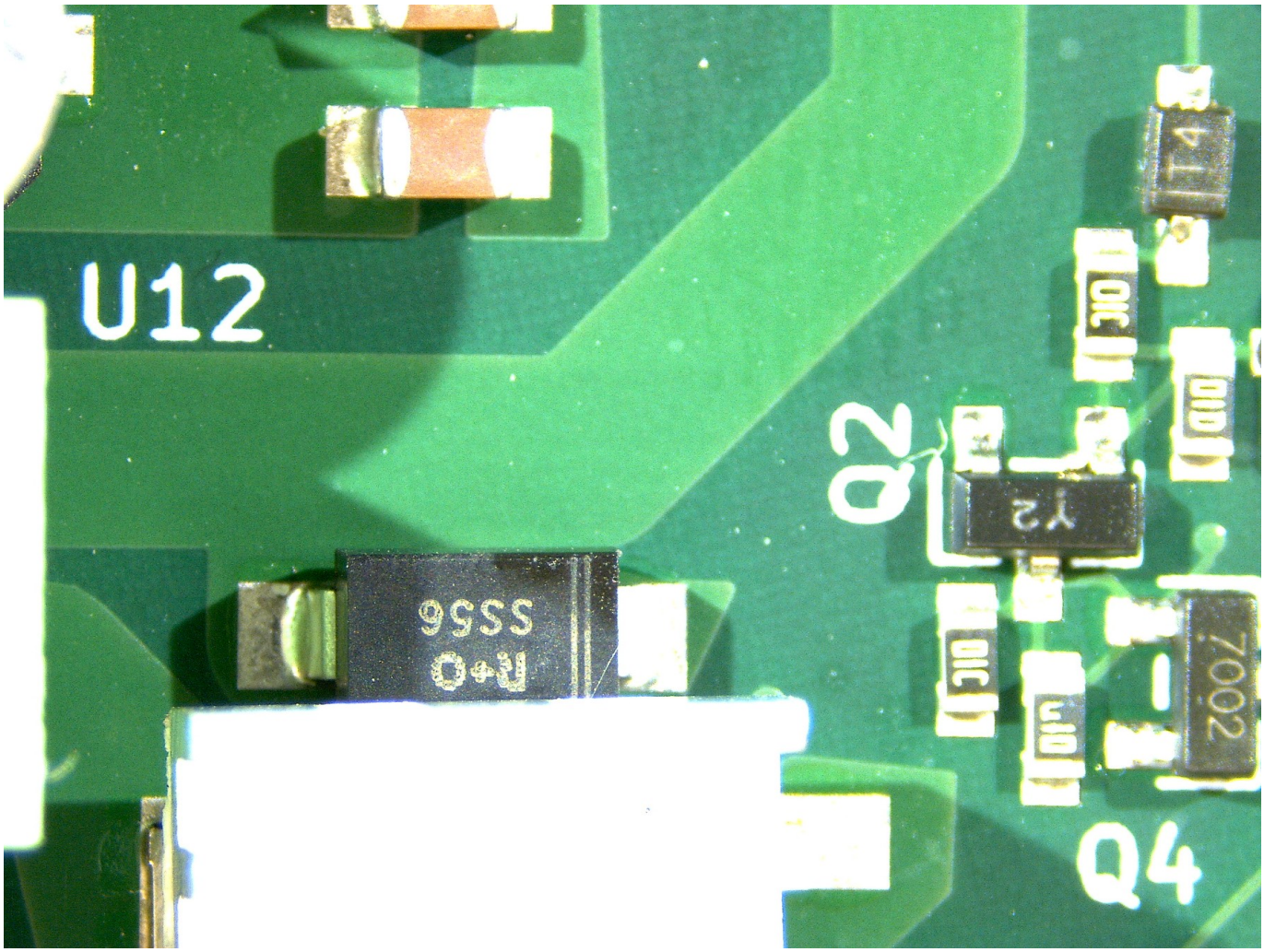
D18

Submitted orientation:

- `D18` pad 1 (left, in this image) is `heater_hi`.
- `D18` pad 2 is `heater_lo`.
- The submitted part wrapper maps pin 1 to cathode and pin 2 to anode.

Observed:

- The physical stripe is on the right side, closer to `Q2`.
- That is the `heater_lo` side, opposite the submitted orientation.



D18 microscope photo

Relevant Files

If the submitted design files are not currently available on your end, the frozen copy of the release I submitted is available on GitHub:

- Frozen submitted release folder
- Submitted KiCad board file
- Submitted JLC BOM
- Submitted JLC CPL / centroid file

Ideal Outcome For Me

I would prefer a refund or JLCPCB account credit rather than a remake of the same revision, because I am preparing an updated board revision with clearer polarity/orientation markings before placing the next order.

I appreciate you taking the time to review this, please let me know if you have any questions or need any additional information!

Best,

Riley